AMD DEVELOPER HACKATHON · ACT III · INTELLIGENT INDUSTRY

SIMON

Verifiable LLM inference on hardware that you do not control

Nodes sign what they computed. Anyone can verify it offline. Live on Gemma 4.

github.com/Szpili/SIMON · MIT · Rust · libp2p

THE PROBLEM

A signature proves who wrote the number.
It never proves the number is true.

Every decentralized inference network hands you a signed claim and calls it a proof. Ours started the same way. The interesting engineering is in the gap between those two words.

What a receipt proves and what it does not

Proved today

A registered node signed this exact answer as the result of this job.

- Ed25519 signature over a commitment to the output text
- Bound to the job id. A valid receipt from another run cannot be reused

Not proved

That the declared model generated it.

- A malicious node can sign arbitrary text with a correct digest
- Token counts and timings are signed declarations, not independent measurements

Binding model + prompt + execution + output needs an execution audit. It is not implemented, and the repository says so.

LIVE DEMO

Try to forge the receipt yourself

Tamper with the answer

Change one byte after signing.

REJECTED: the signature breaks

Replay another job

A correctly signed receipt from a different job.

REJECTED: not proof of this work

Swap the text

Real signature, substituted answer.

REJECTED: the digest covers the text

No simulated mode. If no node is reachable, the demo says so instead of a canned result.

INTELLIGENT INDUSTRY · LIVE

Maintenance tickets with an audit trail

The maintenance planner of a machining plant turns machine alarms into tickets by hand, late, and cannot later prove which system produced which advice.

1 · Rules decide

Physics thresholds flag the risk and the failure mode. The model never decides it.

2 · Gemma 4 drafts

Gemma 4 26B-A4B writes the technician ticket from the rule findings only.

3 · Checked

JSON, modes equal to the rules, no invented numbers, no order to stop or bypass equipment.

4 · Signed

The GPU node signs the exact ticket. The page re-checks every receipt in the browser.

15/15 tickets passed every check, median 1.5 s each, all receipts valid. Nodes: Gemma 4 on the plant's RTX 3090 and on an AMD Instinct MI300X in AMD Developer Cloud.

pop-os.tailaf1556.ts.net/simon/industry · AI4I 2020 data (UCI, CC BY 4.0, synthetic) · a technician reviews every ticket

MEASURED, NOT PROJECTED

An audit is cheap enough to pay for

An audit is one forward pass. The job must also decode, and decoding is ~55× slower per token.

Prompt	Output	Job	Audit	Audit as % of job
500	50	7.9 s	0.63 s	**8.0%**
2016	200	18.8 s	1.86 s	**9.9%**
2011	500	87.1 s	2.07 s	**2.4%**
6037	500	69.1 s	5.12 s	**7.4%**

Lower bound: this is the forward pass alone, without activation extraction and hashing. Raw data is in the repository.

Heterogeneous on purpose, and it shows

Two GPUs, two serving stacks, two models, one network. The engine and the model on each result come from the signed receipt, not from a label that we wrote beside it.

RTX 3090 · vLLM 0.27.1

Qwen3.8-27B · full context admission gate

64.94 tok/s

RTX 3080 Ti · Ollama 0.34.0

Bielik 11B · admission gate disabled, and the node says so

6.50 tok/s

A node that cannot measure its own context budget declares that. It does not pretend.

An honest AMD node adds no false positives

We tested it on one MI300X on 2026-10-11. In the pre-registered test, an honest AMD node flagged the same 5 of 32 prompts as the NVIDIA reference flagged against itself.

Measured on the MI300X

AMD output was byte-identical to NVIDIA on 22 of 32 prompts. A SIMON node ran Qwen3.8-27B on the MI300X. Its receipts verify, and one changed byte is rejected. One card sustained 64 concurrent streams.

Scope and verdict

One 3B model, Q8, llama.cpp, a 20-token window. The verdict is still FAIL, because honest configurations flag prompts. The M3 check stays telemetry.

We publish the result as measured, including the FAIL. Live now: Gemma 4 on an MI300X node.

MEASURED ON THE MI300X · 2026-10-13

SIMON adds no measurable cost

We sent the same jobs to vLLM directly and through a SIMON node. The client verified every receipt.

Concurrent streams	vLLM direct	Through SIMON	Receipts verified
1	65 tok/s	65 tok/s	**6 of 6**
4	220 tok/s	220 tok/s	**24 of 24**
16	712 tok/s	722 tok/s	**96 of 96**
64	**1948 tok/s**	**2020 tok/s**	**384 of 384**

Qwen3.8-27B BF16, vLLM 0.27.1, 256 tokens per job. One job takes about 14 ms more. The client ran on the same host, so the WAN is not in these numbers.

Two layers check which model ran

A receipt shows who signed the output. These two layers check the model from outside the node, through normal SIMON jobs.

The node serves	Agent behavior (LIDAR-style)	One-word answers (JSD)
another model family	**detected**	**detected**
another size (0.5B, 3B, 7B, 14B)	**detected**	**detected**
the same weights in Q4 instead of Q8	**not detected (0 of 4)**	**detected (0.43 to 0.51, noise up to 0.27)**
the same model on another GPU (honest)	**no false alarm (16 of 16)**	**no false alarm (0.20 to 0.25)**

7 model classes on CPU, GTX TITAN X and RTX 3090. Every agent step was a SIMON job with a verified receipt. 4 to 10 fingerprints per class: a pilot, not an error rate.

Three of our own claims, withdrawn in public

The receipt did not bind the answer

Found during a test of whether we can count tokens at all

The digest covered metadata. The text traveled unsigned.

A constant pretending to be a measurement

It survived the code, a comment that claimed "hard measurement", and a demo that showed one number on two GPUs

Throughput was tokens / 65.0, the figure of one card, applied to every node.

A test that the whitepaper promised and that we never wrote

Corrected in both languages. The ledger now gives it somewhere to be true

Both papers said that the client tests whether a receipt is single-use. It did not.

Status, without rounding up

Capability	State
Output binding	**implemented, tested**
Offline receipt verification by a third party	**implemented, tested**
Chunking, map-reduce, agent-to-agent chaining	**working**
Concurrent jobs on one node, with a busy limit	**working, measured on MI300X**
Which model a node serves (two-layer verifier)	**pilot: family, size, quantization**
Token counts recomputed independently	**partial: backends expose no token ids**
Execution correctness (TopLoc-style audit)	**not implemented**
Automatic slashing	**off by decision: no defensible threshold**
Wash-compute resistance	**unsolved**

192 tests green · CI on every push, with a guard against constants that pose as measurements

Idle capacity turned into burst capacity, with a receipt that you can verify

This is not "earn while you sleep". At 350 W, a night of full load is worth less in commodity tokens than the electricity. The value is work that must not leave your machines, and results that you can verify. It is also five GPUs at once on the morning you need them.

Code, MIT

github.com/Szpili/SIMON

Also in the repository

Whitepaper (EN and PL) · security log · ROCm notebook